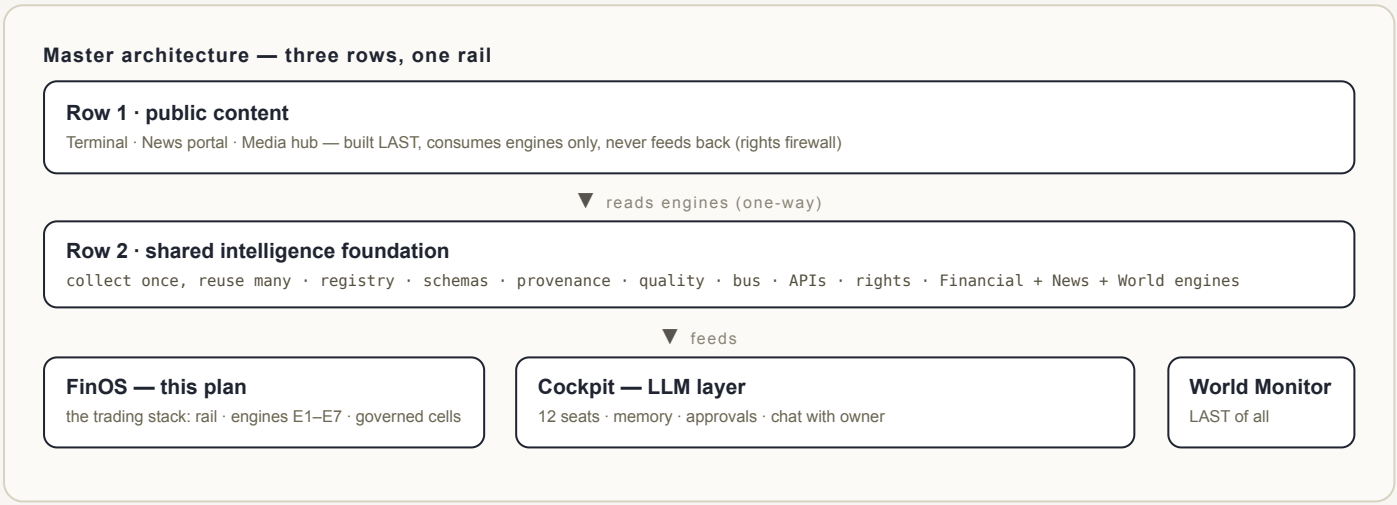
FINOS — the financial trading plan of the whole system, in one picture plane

The trading stack (Row 3 of the ecosystem) end to end: data in → stores → engines → governed execution → ledger, with the LLM cockpit layer wired around it. Every number is a locked decision from Sessions 1–46; labels: ESTIMATE = to measure at its gate · MEASURED = on record.

75	480–545	12	<5 ms
research brains (owner said "73" — old symbols×1-style count; roster source of truth is 75)	executor cells ceiling (built only when E4 promotes a style; ≤75 active at once)	LLM seats in the cockpit — decide-and-explain, never the hot path (R3)	internal order-path budget; venue RTT dominates (R8 honesty)



Non-negotiable invariants printed on every following page

R1 collect once / reuse many · R2 rights firewall at boundary · R3 no LLM in fast paths · R4 PIT + survivorship · R5 provenance everywhere · R6 versioned contracts · R7 train→eval→paper→approve→deploy · R8 venue RTT dominates · G1–G6 fleet guards · L0–L4 kill cascade + out-of-band kill

Data pipeline — in motion · at rest · at insight · at action

Same plane as the reference "Data Pipeline" layout: everything flows over one bus; every store has exactly one job.

Data in motion — collectors rail (R1)

Venue feeds
Upstox (NSE/BSE WS+SSE) · cTrader Open API (FX ECN) · Hyperliquid (perps) · tick+, depth for FX Team 2 + crypto later

News & world
news collectors (B1, W0b–W1) · world structured daily/1-min · T3 pin-to-pin only per research need — never blanket

Reference & filings
bhavcopy, filings, calendars, fundamentals PIT-as-reported (O12 open) — registry rows carry license class + verify state

▼ **validate · as-of stamp · rights-tag — R1 rail (Rust)** single writer to stores

The bus

NATS JetStream — real-time streaming & messaging (D2)
`market.* · signal.* · order.* · risk.* · ledger.* · seat.* — queue-groups = load balance · durable consumers = replay · partitioned per-key ordering`

▼ ▼ ▼ ▼ fan-out

Data at rest — the store ladder

D0 · Parquet lake
years of archive on NVMe + offsite object (16-TB target accepted S43) · research scans, replay, backfills

D1 · QuestDB (warm)
live window only · E1 pre-warm, PIT replay · final §19; ClickHouse fallback if W0a gate fails twice

D3 · Postgres (reference)
registry · schemas · provenance · rights classes · symbols · calendars · cost model · limits · llm_tasks

D4 · Postgres (ledger)
append-only role · fills / P&L / slippage / G4 trials · ≥5 yr (R13) · no UPDATE/DELETE grants

▼ nightly D0/D1 → D6 · research reads columnar only

Data at insight — offline (never the hot path)

D6 · DuckDB research scans
vectorized replay for Mode R · one process rule

E4 gate stack
E4-SIM swarm (10^4 worlds) · E4-BT vectorized + event · E4-WF CPCV · DSR ≥ 0.95 · crisis replay

Scoring + research seats
cited artifacts (quant-mind contract) · D5 candidate prep · 10^3 candidates/style cap

▼ owner approval ≤5/week (C8) → D5 signed

Data at action

E5 governor
deterministic: stops · size caps · news ladder · kill-switch

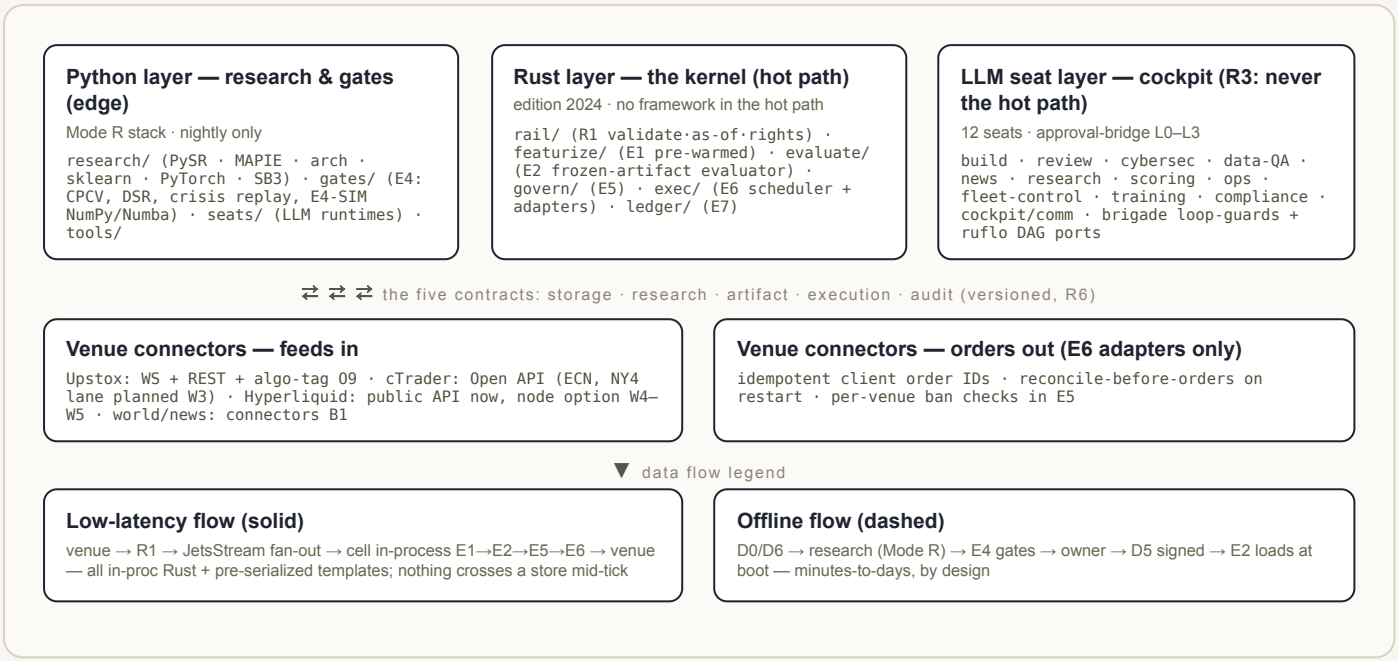
E6 execution
PAPER · SHADOW (zero order authority) · LIVE adapters

E7 ledger
fills vs expected · D4 async writes

Cockpit
L0–L3 approvals · monitor surfaces · chat

Three runtime layers — Rust kernel · Python edge · LLM seat layer

Same plane as the reference "Python/Rust/C++" layout, mapped to our locked parts. Contracts (§3.3 of the blueprint) are the only crossings.



Adopt-vs-build, one line each

Build (no OSS substitute)	R1 rail · E1 · E2 · E5 · E6 adapters · E7 + cost model · D5/G4 · PIT replay · rights firewall · cockpit layer
Adopt (license-checked)	Parquet · QuestDB · NATS · Postgres · DuckDB · PySR · MAPIE · arch · sklearn/PyTorch/SB3 · Nautilus behind the E6 interface · Graphify (local tool)
Patterns ported from the 13 repos	brigade loop-guards/approval/cron · ruflo DAG · agency personas · day-trader checks · quant-mind citation schema · Vibe anti-fabrication · timecopilot stub-gate

Execution path with latency bands — hot · warm · cold

Same plane as the reference hot/warm/cold trading layout. Latency budgets are ESTIMATE until each gate measures them into D3 dashboards.

HOT <1 ms decision · ≤ ~2 ms R1→cell fan-out P99 (W0b)

WARM ms-class D1 reads — boot/replay/dashboards only

COLD seconds–minutes research · seat tasks async by design

Market data in — (~ms network)

Upstox · NSE/BSE
WS + SSE · options TICKER only

cTrader · ECN
7 pairs + 4 commodities · scalping feed = tick+depth

Hyperliquid
BTC/ETH/HYPE perps

News / world rail (B1)
async to the tick path

▼ R1 rail: validate · as-of · rights-tag

HOT PATH — trading EC2-class kernel
Risk first, always:
E5 governor — pre-trade size/limits/spread/news-ladder · E1 features (pre-warmed RAM) · E2 frozen-artifact evaluator · circuit daily kill-switch

WARM PATH — OMS-class
E6 order state machine · lifecycle · ephemeral-replay · position reconciler (reconcile-before-orders invariant) · D4 position ledger (post-settlement writes)

→ order events ▼ E1→E5→E6→wire <5 ms internal (R8); venue RTT dominates

Venues — with RTT labels (vendor-claims labeled)

Upstox
~45 ms round trip (vendor-claim, unaware) · algo-tag digit pending O9

cTrader GA-routed
NY4 lane planning at W3 · prod limits O8 (500/min is demo)

Hyperliquid
~600–700 ms public API (MEASURED public figure) · node option at W4–W5

▼ fills async

Settlement of truth — Postgres / QuestDB + off-graph stores

D4 append-only
fills · slippage · G4 trials · audit — the only history that matters

D1 QuestDB
warm window · PIT replay evidence

COLD PATH — overnight
G4 research + strategy library · E3 training (Colab T4 verified lane) · feature store + model registry in D5

Streaming platform — sources → bus → consumers → outputs

Same plane as the reference "Event/Data Streaming Platform". Exactly five transports in the whole system — anything else is a design bug.

Data sources → producers

Sources
`venue feeds · filings/statistics (stats/reporting) · news endpoints · world datasets (datasets CDC-style refresh) · APIs/webhooks`

Producers = B1 collectors
one row of the 1,114-row TODO each · producer-N scale-out pattern · every publish as-of+rights stamped

▼ API gateway analog: cockpit L0–L3 approval bridge (auth before any action that matters)

Core streaming platform — NATS JetStream (D2)

Message bus
`JetStream durable + at-least-once + exactly-once de-dup on client IDs · subject tree market.*/signal.*/order.*/risk.*/ledger.*/seat.*`

Event store / log
durable streams = replayable history · every cell can rebuild state from the log

Schema registry analog
the five contracts (versioned R6) — enforced at publish, not by trust

Security layer
rights firewall at the R1 boundary · no secrets in configs · append-only DB roles

Observability
`per-hop metrics · flight-recorder traces · G2 dashboards · G9 token burn`

▼ subscribe / queue-groups

Consumers

Mode-T cells (the hands)
≤75 active · one per symbol × style

LLM seats
seat.* subjects + llm_tasks queue

E7 ledger·dashboards
D4 writer · G2 surfaces

▼

Destinations

D0/D1 stores
archive + warm

Cockpit
chat · approvals · monitor

Owner mobile
out-of-band kill + status

Digest → vault
weekly auto-write

The five transports (and nothing else)

1 · NATS subjects	the live wire — µs–ms fan-out, load-balance, durable replay
2 · Files	D0/D5 — single-writer discipline
3 · SQL	D1/D3/D4 concurrent servers (headroom = G1 bench)
4 · Embedded DuckDB	single D6 process — never critical path
5 · In-proc calls	E1/E4 workers — µs-class, inside one kernel process

The engine stack E1–E7 — and the candidate ladder

A candidate rides the same ladder at every rung with the same signed artifact and the same E1 code — bit-for-bit, or it doesn't move.

E1 · featurize
bars · ticks · Greeks · indicators · microstructure — Rust, pre-warmed RAM at boot

E2 · evaluate
frozen-artifact evaluator each tick · compiled formulas / ONNX

E5 · govern
zero ML — deterministic or nothing

E3 · train (offline)
SB3/PyTorch wrapper · nightly · Colab T4 verified lane

E4 · THE gate
E4-SIM 10^4 worlds · E4-BT vectorized+event · E4-WF CPCV · DSR ≥ 0.95 · crisis replay

E6 · execute
PAPER / SHADOW (zero order authority) / LIVE adapters — the only engine that ever sends an order

E7 · ledger
fills vs expected · slippage curves → D4

One candidate's life — top to bottom

1 · E4-SIM	10^4 deterministic simulated worlds · seed + param-hash replay · minutes
▼	
2 · E4-BT vectorized	fast screening over D6/DuckDB · seconds
▼	
3 · E4-BT event-driven	fill model: feed latency, order latency, queue position (hftbacktest checklist)
▼	
4 · E4-WF walk-forward	CPCV purge = label horizon + embargo · day-grouped folds across symbols · untouched final holdout · DSR ≥ 0.95 vs decorrelated trials · crisis replay (EOD index-option history)
▼	
5 · E6-PAPER	real hot path, simulated fills · days
▼	
6 · E6-SHADOW	real data · zero order authority (physically) · 3–6 months
▼	
7 · Owner gate	≤5 approvals/week (C8) · L2/L3 card in the cockpit
▼	
8 · D5 signed → LIVE	artifact signed into D5 · Mode T loads at next boot · E7 watches decay
▼	
9 · Monitor / retrain / retire	rolling IC · PSI/edge decay triggers retrain — never raw PSI (C9) · retirement = per-cell, owner-approved, G4 lineage

The workforces — fleet and cockpit, and how they loop together

"One team of agents": the deterministic fleet executes and proves; the LLM seats decide and explain; the gates say who goes live.

The non-LLM fleet (the hands)

Stocks ≤ 59
NIFTY + SENSEX + optionable weight names · options BUYING only, same-day · 8–9 styles/symbol · rule ladder 1–6 + kill

FX Team 1 · 11 symbols
3 styles/pair (session · carry · news) · cascade-flatten · weekend-gap · funding-window · kill

FX Team 2 · 2 symbols
XAU/USD + WTI · pure scalping push-mode · separate account · session-closed flat

Crypto · 3 symbols
funding · liquidation · weekend · PAIR stat-arb · 5-liquidity-gate exit · kill

▲ ▼ state up · commands down — NATS subjects + D3 cell-registry

The LLM cockpit (12 seats — all R3)

build — code + tests
review — PR gate
cybersec — egress · secrets
data-QA — gap tickets
news — digest + ladder
research — citations required
scoring — E2 edge-lift gate
ops — incident narration
fleet-ctl — state map
training — G4 runs

▲ ▼ findings/PRs/approval cards — never market writes outside L0–L3, never frozen-artifact edits without E4

The ten operating loops (the heartbeat)

```
build · fix-bug · promote · research · news · data-QA · ops/kill · train · monitor · self-learn — each loop: trigger → seat → fleet action → gated output; nothing a seat says becomes live without a gate (E4/E2, DSR, L0–L3)
```

Swarm agents — both kinds

Fleet swarm	E4-SIM: 10^4 deterministic worlds for any strategy question — vectorized NumPy/Numba, seed+param-hash replay
LLM crew swarm	multi-seat missions on a ruflo-port DAG — crew.yaml: roster · depends_on · tool whitelist · budget cap · approval lid · CI-linted
Cross-cell swarms	any new style tested against existing cells' exposure in sim before a merge lands

Every agent's memory — and the control plane that keeps it safe

Write authority is enforced at the DB-role level, not by prose. The kill always works without the app.

Memory matrix — fleet and seats side by side

Fleet (75 brains / 480–545 hands)

working → cell process state (tick/feature/position/DD)
episodic → E7 ledger + G4 trial log + shadow history
semantic → R3 registry: frozen artifacts · calendars · limits · ban status · regime→style map
procedural → the code + runbooks (git PR only)

LLM seats (12)

working → task lease payload in llm_tasks
episodic → D4 audit rows (tokens · result · artifacts)
semantic → D5 artifacts + Graphify graph + vault notes
procedural → skill files (PR + review)
owner rules → vault note (cockpit seat writes only)

Control plane — approvals and the kill cascade

L4 · global — kill all

L3 · market — kill one market

L2 · team/venue — one account

L1 · symbol — one instrument

L0 · cell — one executor

Owner tiers
L0 chat answers · L1 sandbox checks · L2 sign-offs (ladder, budget, triage) · L3 live decisions (promote, kill re-arm, deploy — approval cards in chat)

▼ cascades down · the out-of-band watchdog + venue-side max-loss order + one-action mobile kill sit OUTSIDE the app

Safety net checked monthly (chaos drills)

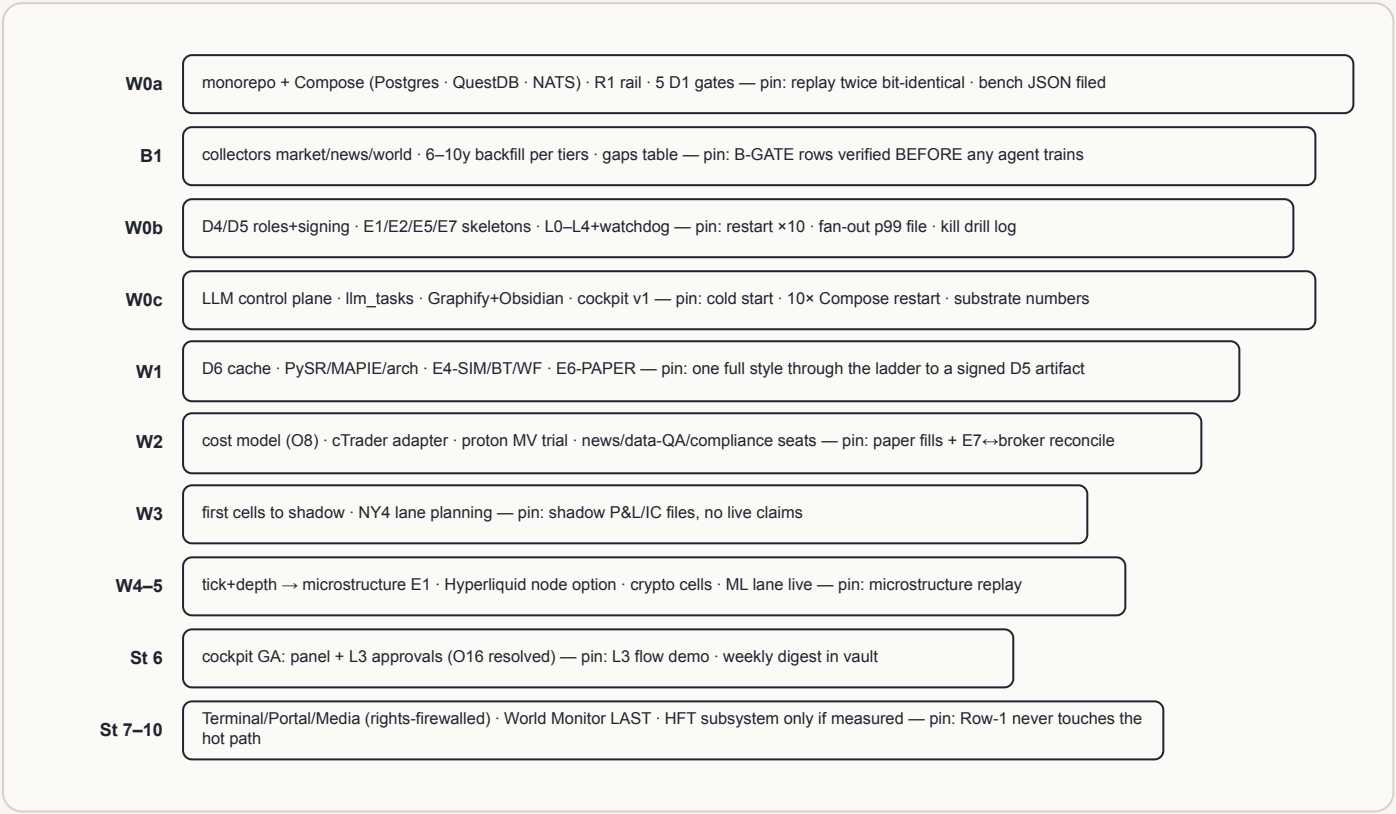
10× restart · feed-stale · venue-down · kill-switch drill · restore from Compose snapshot · reconcile-before-orders on every boot

Cockpit monitor surfaces (G2) — one line each

Cell health	heartbeat · state · last decision id · kill status per cell
Portfolio	D4-derived exposure/net across the 3 sub-accounts
Market state	session clock per venue · kill-state · ban status
Data	gap calendar · 16-TB tracker · collector backlog
LLM layer	token burn/seat/day (G9) · eval replays · failed tasks
Research	G4 trial counts · DSR status · coverage-map heat
Ops	incident queue · drill results · backup freshness

Build order — waves with exit pins that must print real output

Data first → agents second → history grows third (owner order, S42). A wave is DONE only when its pin commands produce real, filed output.



Week-1 concrete instructions

- Create the monorepo skeleton (§2.1 of the blueprint) and bring up Compose with healthchecks.
- Write the R1 rail minimal loop: validate → stamp → publish to market.* → dump D0 parquet.
- Build the replay-harness CLI and the G1 concurrency-bench stub — these two ARE the W0a gates.
- Import the 5 plans + this blueprint into mani/docs · create the vault note types.
- Build the Graphify baseline graph (--update) · open the W0a gate document — every measurement lands there as it happens.

Honesty register — measured vs. estimated vs. rejected

The plan contains zero unverifiable performance claims. Every number keeps its label from the source session.

Vendor-claims and author-claims — labeled, never repeated as fact

```
Upstox ~45 ms RTT · "80% token-cut" (memory-setup, unreproducible by its own README) · Any "71.5×" claim → replaced by our own W0c
fit-test · Graphify Cloud upsell REJECTED
```

What we measure ourselves, at its own gate

```
W0a: 5 D1 gates (burst · determinism · pre-warm · restore · 10× restart)
W0b: fan-out P99 + crash-restart ×10
W0c: Graphify recall delta + recall-substrate shootout + cockpit cold start
W1: DSR ≥ 0.95 + crisis replay + 10³-candidate cap
W2: ingest parity for proton MVs vs JetStream benchmark
```

Rejects on the record (why they were excluded)

AI-Trader	no LICENSE file (badge links to nothing) · hosted platform, not a library
nixtla SDK	cloud-locked api.nixtla.io · the free models live in statsforecast/mlforecast
laya-mlx runtime	MLX is darwin/arm64-only, CUDA unsupported in code · decision-head contract adopted instead
OpenStock code	AGPL-3.0 — UX ideas only, zero code lift
TimescaleDB for D1	tsl/ license wall verified upstream · QuestDB is D1 (§19 final)

Open decisions carried to their waves

O3 F&O granularity · **O4 off-index names** · **O5 first style** · **O6 FX scalping add** · **O7 hedge executor** · **O8 cTrader prod limits** · **O9 Upstox algo-tag** · **O10 order-balance def** · **O11 HL FIU-IND/FEMA/tax** · **O12 PIT fundamentals** · **O13 ToS checklist** · **O14 model routing (W0c)** · **O15 NSE membership vendor (W2)** · **O16 cockpit runtime (W0c)**

Sources: FULL-SYSTEM-BLUEPRINT (S46) · FULL-SYSTEM-MASTER-PLAN (S45) · AGENT-FLEET-BLUEPRINT · ENGINE-DATABASE-PLAN (incl. §19) · LLM-FLEET-BLUEPRINT · planning log S1–S46. This document adds no new external claims. The reference architecture images inspired layout/style only; all engineering choices are our locked decisions.